

How Much Can Be Torn Away? Measuring the Limits of GenAI-Based Repair of Damaged Historical Newspaper Text

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Abstract

[TODO: Write abstract after results are final.]

1 Introduction

Digitised newspapers are now a core source for historical research. Keyword search over large digital collections has replaced much of the slow reading of physical volumes (Bingham, 2010). Yet what historians search and mine is not the page itself but the text that an OCR engine read from a scan of it. OCR errors measurably change the results of quantitative analyses (Hill and Hengchen, 2019) and harm downstream NLP tasks, with named entities among the most affected (van Strien et al., 2020; Hamdi et al., 2023). For 19th-century British newspapers the mean character error rate (CER) is about 7%, but it ranges from 0.3% to 44.5% between excerpts of a single title (Booth et al., 2024). This spread tracks the condition of the source page, not just the OCR engine: torn, stained or faded originals give the engine the least to work with (Holley, 2009). For the British Library’s own 19th-century newspaper archive, character accuracy reaches 83.6% but word accuracy falls to 78%, and accuracy for capitalised words and number groups drops further still, to 63.4% and 64.1% (Tanner et al., 2009). The words hit hardest are thus often the facts a historian searches for: names, places, dates and sums of money.

Generative AI makes it tempting to repair such text automatically, but the evidence is mixed. Llama 2, instruction-tuned on BLN600, reduced the CER of its OCR by 54.5%, against 23.3% for a fine-tuned BART (Thomas et al., 2024), while a study of 14 LLMs on several historical benchmarks found them “anything but efficient” at post-correction (Boros et al., 2024). There is also a deeper risk. A generative model always returns fluent text, even where the evidence for a word is gone, and such models are known to produce content that

their input does not support (Ji et al., 2023). For an archive, a plausible but wrong name is worse than visible noise, because the reader can no longer see that anything was damaged. Similarity-based metrics may hide this error too: a sentence with the wrong name can still score as close in meaning to the original.

Existing post-OCR studies score whole documents at whatever error level the OCR happened to produce. They show that a model helps on average, but not how far repair can be trusted as damage grows, nor whether the facts inside the damaged region survive. We therefore ask:

RQ. When the damaged words of a sentence are known and their garbled letters are visible, how much of the original wording, and in particular its names, numbers and dates, can a fine-tuned masked language model and a few-shot instruction-following LLM restore, and how does this change as the damaged share of the sentence grows?

We split it into three sub-questions. **SQ1** (severity): does repair quality fall smoothly as damage grows, or break down at some level? **SQ2** (method): given identical input, does a fine-tuned masked language model or a few-shot LLM repair better? **SQ3** (metric validity): does semantic similarity overstate repair quality compared with exact fact recovery?

Answering these questions needs damage whose amount we control and whose original text we know, which real damaged pages cannot give. We therefore build a controlled experiment on BLN600 (Booth et al., 2024), a corpus of 600 newspaper excerpts with human gold transcriptions. In each of 150 test sentences we damage one block of words around a fact at five nested levels, from one word to 75% of the sentence, with character noise calibrated on the corpus’s own OCR errors. Both methods receive the same input: the intact

Level	1w	10%	25%	50%	75%
Sentence CER (%)	1.0	3.2	7.9	15.5	23.1
Facts in span (mean)	1.0	1.6	2.2	3.2	4.2
Facts / damaged words (%)	100	48	26	19	17

Table 1: Damage by level (test split). CER is over the whole sentence against gold; real BLN600 OCR is about 7% (Booth et al., 2024).

context plus the garbled form of each damaged word, so that only the repair differs. We compare two GenAI families: [TODO: CHECK IS BERT GenAI?] masked infilling with BERT (Devlin et al., 2019), fine-tuned on in-domain gold text and guided by letter similarity in the spirit of “context beats confusion” (Evershed and Fitch, 2014), and instruction-following generation with an open-weight LLM prompted with three solved examples. We score the repaired words with BERTScore (Zhang et al., 2020) for meaning and with a Fact Recovery Rate for exact facts, and we probe the LLM for memorised test text (Sainz et al., 2023). Repair here works on text only: we simulate the OCR output that damage produces, not the physical page, and we assume the damaged region is known; detecting it, and repairing directly from the scanned image with a vision-capable model, are both left to future work.

Our contributions are (i) a fact-centred, graded damage benchmark on BLN600 with nested spans, so the same place gets worse from level to level; (ii) a controlled comparison of masked infilling and few-shot generation under identical input and scoring; and (iii) a fact-level metric next to BERTScore that shows where semantic similarity and factual accuracy part ways. [TODO: Key outcomes, from the TEST split only: (1) the level at which Fact Recovery Rate falls to the no-repair floor; (2) which method wins at low vs high damage, and whether a crossing is a format-failure effect; (3) the size of the BERTScore vs Fact Recovery gap.]

2 Related Work

[TODO: Draft related work.]

3 Method

3.1 Task and Data

[TODO: Draft 3.1 (about 8 lines).]

3.2 Damage Simulation

[TODO: Draft 3.2 (about 10 lines).]

<i>BERT + reranking</i>	
Base model	bert-base-cased (110M)
Fine-tuning	lr 5e−5, batch 16, early stop (patience 2, max 10 ep.), best epoch 2
Val. loss	5.61 (pretrained) → 4.14
Reranking	1–3 masks/slot, beam 5, top 10, $\lambda = 8$
<i>LLM few-shot</i>	
Model	[TODO: final id]
Prompt	system msg + 3 dev examples, JSON output
Decoding	temperature 0, max 1024 tokens

Table 2: Implementation settings. λ chosen on dev from {1, 2, 4, 8, 16}.

Factor	Levels
Severity	0, 1 word, 10, 25, 50, 75% of words
Length band	20–29, 30–44, 45–60 words
Methods	BERT + rerank; LLM few-shot; no repair (secondary: LLM + article)
Data	150 test sentences, 750 rows per method; dev 20 (tuning only)
Metrics	BERTScore, Fact Recovery Rate (primary); exact match, repair gain, format failures, contamination (secondary)

Table 3: Experimental design.

3.3 Shared Slot Input

[TODO: Draft 3.3 (about 5 lines) and Figure 1 in figures/.]

3.4 Repair Methods

[TODO: Draft 3.4 (about 18 lines).]

4 Empirical Investigation

4.1 Experimental Design

[TODO: Draft 4.1 (about 6 lines).]

4.2 Metrics

[TODO: Draft 4.2 (about 8 lines).]

4.3 Results

[TODO: Fill after the LLM test run and notebook 04 on test.]

5 Interpretation and Conclusion

[TODO: Draft after results.]

Limitations

[TODO: Draft limitations as one dense paragraph or a tight list.]

Level	BERTScore (span)			Fact Recovery Rate		
	BERT	LLM	None	BERT	LLM	None
1w	—	—	—	—	—	—
10%	—	—	—	—	—	—
25%	—	—	—	—	—	—
50%	—	—	—	—	—	—
75%	—	—	—	—	—	—

Table 4: Test results, 150 sentences per level. None = no repair. [TODO: LLM format failures per level; paired counts.]

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Appendix Outline (for the defence, not part of this report)

[TODO: One short paragraph listing these items.]